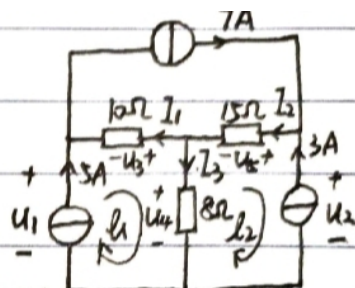


5、6、7、8课时作业答案

1.9

解:
$$\begin{cases} 7A - 5A - I_1 = 0 \\ I_2 - 7A - 3A = 0 \\ I_1 + I_3 - I_2 = 0 \end{cases} \quad \text{解得} \quad \begin{cases} I_1 = 2A \\ I_2 = 10A \\ I_3 = 8A \end{cases}$$

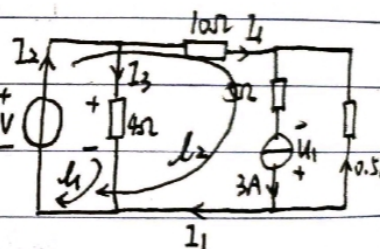


$$\begin{cases} -U_1 - I_1 \cdot 10\Omega + I_3 \cdot 8\Omega = 0 \\ U_2 - I_2 \cdot 15\Omega - I_3 \cdot 8\Omega = 0 \end{cases} \quad \text{解得} \quad \begin{cases} U_1 = 44V \\ U_2 = 214V \end{cases}$$

1.14

1.14解:
$$\begin{cases} I_1 + 0.5A - 3A = 0 \\ I_2 - I_1 - I_3 = 0 \\ I_3 \cdot 4\Omega - 8V = 0 \\ I_1 \cdot 10\Omega + 3A \cdot 5\Omega - U_1 - 8V = 0 \end{cases}$$

解得
$$\begin{cases} I_1 = 2.5A \\ I_2 = 4.5A \\ I_3 = 2A \\ U_1 = 32V \end{cases}$$



故电压源 $P_1 = 8V \cdot I_2 = 36W$, 电流源 $P_2 = U_1 \cdot 3A = 96W$.

1.17

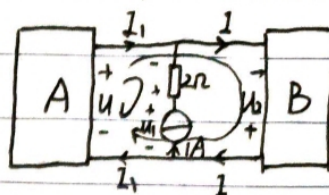
1.17解:
$$\begin{cases} I - I_1 - 1A = 0 \\ -U - U_2 = 0 \end{cases} \quad \text{解得} \quad \begin{cases} I_1 = 1A \\ U_2 = -10V \end{cases}$$

$P_A = U I_1 = 10W$, $P_B = U_2 I = -20W$.

$-U - 1A \cdot 2\Omega + U_1 = 0 \Rightarrow U_1 = 12V$.

$P_{is} = U_1 \cdot 1A = 12W$.

$U = 10V, I = 2A$.

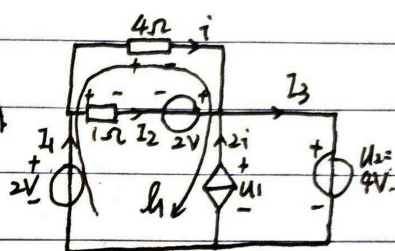


1.18

1.18解:
$$\begin{cases} U_1 = U_2 = 4V \\ -2V + 4\Omega \cdot i + U_1 = 0 \end{cases}$$

解得 $i = -0.5A$

故受控源 $P = U_1 \cdot 2i = -4W$.



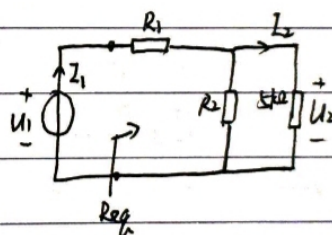
2.4

2.

2.4 解: $R_{eq} = R_1 + R_2 // 5k\Omega = R_1 + \frac{R_2 \cdot 5k\Omega}{R_2 + 5k\Omega} = 40k\Omega$ ①

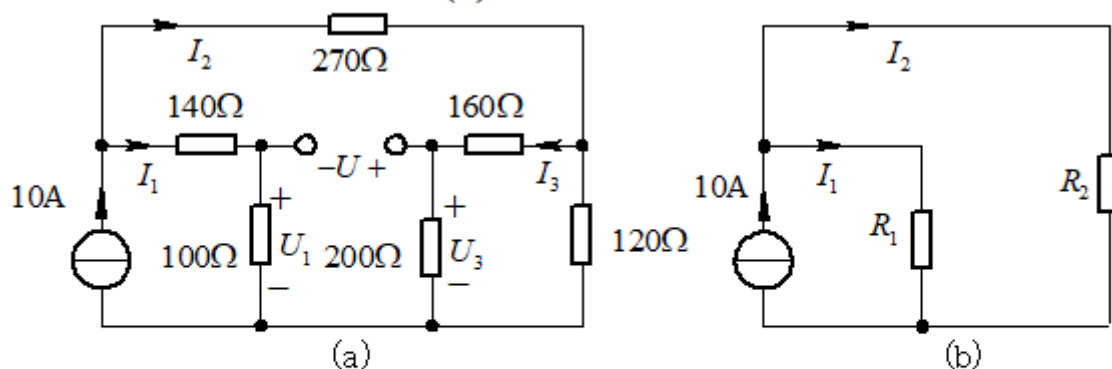
由 $\frac{U_2}{U_1} = 0.05$ 得 $\frac{R_2 // 5k\Omega}{R_1} = \frac{1}{19}$ ②

由①②联立解得 $\begin{cases} R_1 = 38k\Omega \\ R_2 = \frac{10}{3}k\Omega \end{cases}$



6

解: 首先将电路化简成图(b)。



图中

$$R_1 = (140 + 100)\Omega = 240\Omega$$

$$R_2 = \left[270 + \frac{(200 + 160) \times 120}{(200 + 160) + 120} \right] \Omega = 360\Omega$$

由并联电路分流公式得

$$I_1 = 10A \times \frac{R_2}{R_1 + R_2} = 6A$$

及

$$I_2 = 10 - I_1 = 4A$$

再由图(a)得

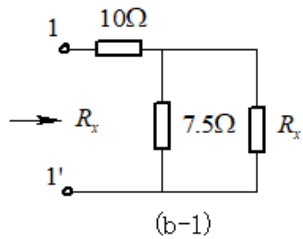
$$I_3 = I_2 \times \frac{120}{360 + 120} = 1A$$

由 KVL 得,

$$U = U_3 - U_1 = 200I_3 - 100I_1 = -400V$$

2.7 (b)

(b) 图(b)为无限长链形电路, 所以从11'和22'向右看进去的等效电阻均为 R_x , 故计算 R_x 的等效电路如图(b-1)所示。参照图(a-1)及式(1)得:



$$R_x = (R + \sqrt{R^2 + 4Rr})/2$$

代入数据得：

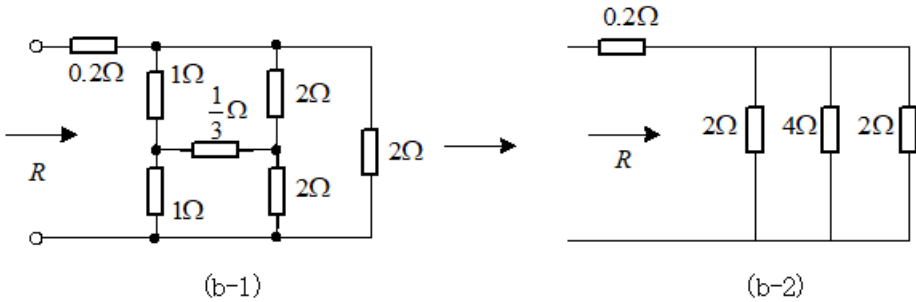
$$R_x = \frac{10 + \sqrt{10^2 + 4 \times 10 \times 7.5}}{2} \Omega = 15 \Omega$$

所以

$$R_x = 15\Omega$$

2.9(b)

(b) 对图(b)电路, 将 6Ω 和 3Ω 并联等效为 2Ω , 2Ω 和 2Ω 并联等效为 1Ω , 4Ω 和 4Ω 并联等效为 2Ω , 得图(b-1)所示等效电路:

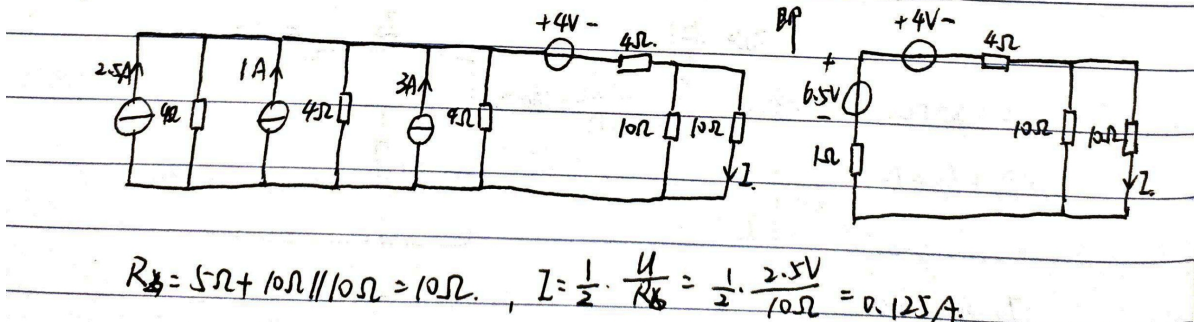


在图(b-1)中有一平衡电桥，去掉桥 $(1/3)\Omega$ 的电阻，再等效成图(b-2)，易求得

$$R = \left(0.2 + \frac{1}{\frac{1}{2} + \frac{1}{4} + \frac{1}{2}} \right) \Omega = 1\Omega$$

2.10

2.10 解: 原图等效为



$$R_{\text{eq}} = 5\Omega + 10\Omega \parallel 10\Omega = 10\Omega, \quad I = \frac{1}{2} \cdot \frac{U}{R_{\text{eq}}} = \frac{1}{2} \cdot \frac{2.5\text{V}}{10\Omega} = 0.125\text{A}$$